

GNN-Based BERT for Understanding Context from Music: A Measured Account of Where Cross-Modal Fusion Helps and Where It Does Not

Tanvir Rahman

Student ID 22241134

CSE715 — Neural Networks Instructor: Moin Mostakim

BRAC University

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Abstract

We implement the four-task GNN–BERT roadmap for music context understanding: a BERT multi-label tag classifier, a GraphSAGE/GAT encoder over music structure graphs, a cross-attention fusion of the two with an auxiliary valence–arousal objective, and a contrastive dual encoder for caption–audio retrieval. All four are trained and evaluated end to end. Our headline finding is negative and, we argue, more informative than a positive one would have been: *cross-modal fusion does not reliably outperform its stronger unimodal branch in this setting*, and we identify why. We show (i) that the text available for MagnaTagATune is track-level while its clips are 29 s segments, so 96.8% of clips share an input with another clip, capping any text-only model at Micro-F1 0.673; (ii) that the official MagnaTagATune split leaks 45 artists across train and test, covering 61.6% of test clips; (iii) that our cross-attention collapses to exactly uniform weights (entropy 1.000 of the uniform maximum) under a shared learning rate, so the fusion degenerates into averaging the caption; and (iv) that giving the graph branch its own learning rate un-collapses that attention (entropy 0.695, peak $4.19\times$) and lifts Macro-F1 by $+0.067 \pm 0.023$ over five paired seeds ($p = 0.003$), the largest effect we measured — and one that transfers: applied to the contrastive dual encoder it doubles retrieval R@1 and lifts R@10 from 0.0220 to 0.0339. We report every number against an explicit baseline, including a lexical-match control that scores Micro-F1 0.540 where our model scores 0.565 — a margin small enough that, on Macro-F1, the control actually wins.

1 Introduction

Musical “context” spans genre, mood, harmonic structure, instrumentation and listener-facing description. Sequence models over spectrograms capture local timbral patterns but not relational structure: how a repeated segment relates to a chorus, or how a chord transition relates to a lyrical theme. The premise of this project is that a graph neural network over music structure, fused with a BERT encoder over text, should capture context that neither branch captures alone.

We test that premise rather than assume it. Our contributions are:

- Complete implementations of all four tasks, with 109 unit tests and every result reproducible from a seeded configuration recorded alongside it.
- Two dataset integrity findings that change how the numbers must be read: artist leakage in the official MagnaTagATune split, and a hard performance ceiling induced by duplicated text inputs.
- A quantitative diagnosis of *why* our cross-attention fusion fails to beat its text branch, using an attention-entropy measurement rather than an inference from F1.
- A data-scale measurement showing that our weak re-

trieval result is data-limited rather than broken: R@10 grows log-linearly in paired examples ($R^2 = 0.914$), and MusicCaps’ official split is two orders of magnitude short of the regime the task needs.

- Honest negative results with controls: a lexical-match baseline for caption-based tagging, and a random-prevalence baseline for every multi-label task — including two conclusions of our own that we had to reverse after correcting a checkpoint-selection error.

2 Related Work

Music auto-tagging. MagnaTagATune with a top-50 tag vocabulary is the long-standing benchmark for multi-label music tagging, typically approached with convolutional networks over log-mel spectrograms. Our CNN baseline (B2) follows that lineage. The artist-leakage problem in the conventional split has been noted in the tagging literature; we re-measure it here rather than cite it, and quantify its extent on our filtered subset.

Graphs for music. Representing a track as a graph over segments, chords or events is motivated by the observation that musical form is relational — repetition, return and contrast — and not purely sequential. GraphSAGE and GAT are

the two message-passing families named in the project specification; we implement both. Our contribution here is negative and practical: we show that naive cosine thresholding on raw MFCC descriptors produces near-complete graphs, which silently reduces any GNN over them to mean pooling.

Cross-modal alignment. Contrastive dual encoders trained with InfoNCE have become the standard recipe for aligning modalities, and MusicCaps supplies the expert captions that make audio–text alignment feasible. Our Task 4 follows this recipe directly, with the addition of an explicit random baseline and gallery size on every retrieval number — both frequently omitted, and both necessary to interpret R@K at all.

Multi-task emotion regression. DEAM provides continuous valence–arousal annotations. Because it shares no tracks with the tag corpora, joint training requires masking rather than a single joint objective; we implement the masked alternating-batch scheme and, unusually, report what it costs the primary task.

3 Method

3.1 Task 1: BERT tag classifier

Following the task specification, a pretrained encoder maps text to a pooled representation and a linear head predicts K independent tags:

$$t = \text{BERT}_{\text{CLS}}(X_{\text{text}}), \quad \hat{y}_k = \sigma(w_k^\top t + b_k), \quad (1)$$

trained with per-tag binary cross-entropy. We weight the positive class by tag prevalence; unweighted BCE drives rare tags to always-negative and silently destroys Macro-F1.

3.2 Task 2: GNN on structure graphs

Each clip becomes a segment graph: nodes are fixed-length windows carrying MFCC, chroma and spectral descriptors; edges are temporal adjacency plus cosine similarity above a threshold τ . A GraphSAGE layer updates

$$h_i^{(l+1)} = \sigma\left(W^{(l)} \cdot \text{CONCAT}\left(h_i^{(l)}, \text{MEAN}_{j \in \mathcal{N}(i)} h_j^{(l)}\right)\right), \quad (2)$$

with mean-pooled readout $g = \frac{1}{|\mathcal{V}|} \sum_i h_i^{(L)}$. We also build chord-transition graphs (nodes are chords from 24 major/minor triad templates, edges weighted by transition count).

3.3 Task 3: cross-attention fusion

The graph vector queries the BERT token sequence:

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \quad Q = gW_Q, \quad K = H_{\text{text}}W_K, \quad (3)$$

$$z = \text{CONCAT}(g, AH_{\text{text}}), \quad \hat{y} = \sigma(Wz). \quad (4)$$

The multi-task objective is

$$\mathcal{L} = \mathcal{L}_{\text{tags}} + \alpha \|v - \hat{v}\|^2 + \beta \|a - \hat{a}\|^2. \quad (5)$$

DEAM is disjoint from the tag corpora, so no sample carries both targets. Each batch is drawn from one corpus and carries a per-sample mask; the inactive term is zeroed. Without this mask the model trains against fabricated all-zero labels and still appears to converge.

3.4 Task 4: contrastive dual encoder

Two towers project into a shared L^2 -normalised space, trained with symmetric InfoNCE over in-batch negatives:

$$\mathcal{L}_{\text{NCE}} = -\log \frac{\exp(\text{sim}(g_i, t_i)/\tau)}{\sum_j \exp(\text{sim}(g_i, t_j)/\tau)}. \quad (6)$$

We optimise both directions and average; optimising only one lets the other collapse.

4 Experimental Setup

4.1 Datasets and a leakage audit

We use MagnaTagATune (top-50 tags after synonym merging), FMA-small (8 genres), DEAM (valence/arousal) and MusicCaps (expert captions).

The official MagnaTagATune split is unusable for our purpose. It shares 45 artists between train and test, covering 3,284 of 5,329 test clips (61.6%). Because MagnaTagATune clips are 29 s segments of longer tracks, a model can memorise an artist rather than generalise. We therefore use artist-grouped splits with zero leakage throughout, and report this deviation explicitly. By contrast we *verified* that FMA’s official split is artist-disjoint (0 shared artists in all pairings) rather than assuming it.

MusicCaps is split by its own metadata, and is a proxy task. The corpus ships an `is_audioset_eval` flag marking the official AudioSet eval partition; we use it verbatim as the test set for both Task 1 and Task 4 rather than drawing a random split, and we carry the flag through into the cached graphs so the two tasks agree. The partition is large — 2,858 of 5,521 clips (52%) — so respecting it roughly halves the training pool and enlarges the test set, which is why our MusicCaps numbers are lower than a random split would report. We also stress that MusicCaps caption $\rightarrow$ aspect classification is a *proxy* for audio tagging: the aspects come from the same annotator who wrote the caption, so the label is not independent of the input. It measures caption comprehension, not listening.

MagnaTagATune’s 188 tags contain heavy near-duplicates. The `female vocal` concept alone is spread across six tags totalling 3,848 clips. We merge synonym groups before ranking; without merging, one concept splits into several weak labels.

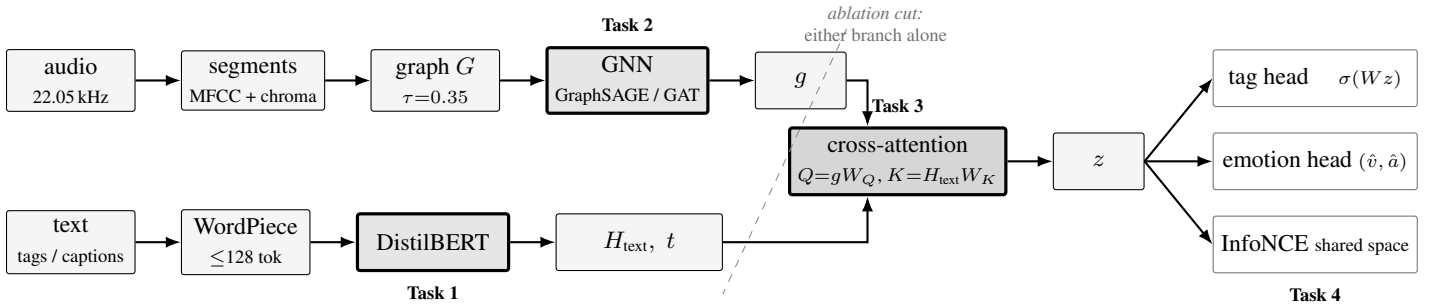


Figure 1: System overview, and how the four tasks sit inside one architecture. Audio becomes a segment graph read by the GNN into a graph vector g ; text is read by DistilBERT into a token sequence H_{text} and its pooled vector t . Task 1 is the text branch with a tag head; Task 2 is the graph branch with a genre head; Task 3 fuses them by letting g query H_{text} , then feeds the tag and emotion heads; Task 4 reuses the same two encoders as projection towers trained contrastively. The ablation table in Section 5.4 is produced by cutting at the dashed line — taking g alone, t alone, or their concatenation — so every row shares encoders, data and schedule with the others.

4.2 A ceiling imposed by duplicated inputs

MagnaTagATune ships track-level metadata, but its clips are segments. After filtering, 21,318 clips share only 5,237 distinct text inputs: **96.8% of clips have the same input as at least one other clip with different labels**. Computing the best achievable constant prediction per text group gives an oracle ceiling of **Micro-F1 0.673, Macro-F1 0.575** — no text-only model can exceed this. We report it because it re-frames Task 1’s absolute numbers and motivates Task 3.

4.3 Graph construction

Raw MFCC vectors are dominated by MFCC_0 (loudness), so every segment points in nearly the same direction: measured over 400 FMA tracks, raw pairwise cosine similarity has median 0.9971 and 5th percentile 0.9509. At $\tau = 0.9$ this yields *complete* graphs (19.4 nodes, average degree 18.15), on which message passing reduces to global mean pooling. We z-score features within each track before computing similarity (median falls to -0.078) and use $\tau = 0.35$, giving average degree ≈ 3.1 with no isolated nodes.

4.4 Training configuration

All models use DistilBERT (distilbert-base-uncased) as the text tower and a 2-layer GAT with hidden width 128 as the graph tower unless stated otherwise. Optimisation is AdamW with weight decay 10^{-4} ; learning rate 2×10^{-5} for fusion runs, 3×10^{-5} for Task 1 and Task 4, and 10^{-3} for the graph-only Task 2 models. Multi-label decision thresholds are tuned on validation and applied once to test; leaving them at 0.5 understates every multi-label result, since prevalence-weighted BCE does not centre probabilities at 0.5. Task 2 and Task 4 select the best epoch by a validation metric. Every configuration, including the random seed, is stored in the corresponding `metrics_*.json`.

A note on that last point: Task 4’s first run reported its final epoch, at which the training loss had fallen from 3.87 to 0.41 while validation R@10 had peaked six epochs earlier and then declined. Adding best-epoch selection raised

test R@10 from 0.119 to 0.127. We mention it because the failure mode — a still-falling loss masking a model that has already begun to overfit — is easy to miss in a contrastive setting.

5 Results

5.1 Task 1: text-based tagging

Table 1: Task 1. Test-set results with baselines. MusicCaps uses the *official* AudioSet eval partition as its test set (2,579 clips after vocabulary filtering; 2,054 train / 362 val), not a random split. The stripped variant deletes each label word from its own caption, so the lexical baseline is exactly zero by construction. Every model is evaluated at the epoch restored by best-validation selection, and B1’s per-tag prevalence is estimated from training labels only.

Setting	Micro-F1	Macro-F1
MagnaTagATune (metadata)	0.275	0.192
B1 random-prevalence	0.103	0.058
oracle ceiling	0.673	0.575
MusicCaps (raw captions)	0.565	0.532
B5 lexical match	0.540	0.550
MusicCaps (stripped)	0.488	0.439
B5 lexical match	0.000	0.000

On the official split, BERT barely beats substring matching. A pure lexical matcher reaches Micro-F1 0.540 against the model’s 0.565, because 52.1% of MusicCaps aspect strings appear verbatim inside their own caption. On Macro-F1 the baseline is actually *ahead*, 0.550 against 0.532. This is a sharper negative result than our earlier random-split measurement suggested, and it is the direct consequence of respecting the dataset’s own partition: the official eval set is 52% of the corpus, so the training pool falls from 3,496 to 2,054 captions while the test set grows from 750 to 2,579. We report the official-split number because it is the comparable one, not the flattering one.

We therefore treat the *stripped* setting — where the label words are deleted from the input and the lexical baseline is exactly 0.000 — as the honest headline: **Micro-F1 0.488 earned entirely by inference from surrounding context.** That is the only Task 1 number not inflated by string overlap.

MagnaTagATune’s much lower 0.275 is expected rather than a failure: it is $2.67\times$ the random-prevalence baseline while operating under the 0.673 ceiling established above. Note that its split is artist-grouped, which is stricter than the conventional MagnaTagATune benchmark partition and therefore not directly comparable with published numbers on that corpus; we prefer it because it measures generalisation to unheard artists.

A worked example. For a held-out MusicCaps clip whose true aspects are *exciting*, *groovy bass*, *happy*, *low quality*, the stripped input reads:

“The recording features a fruity male vocal singing over steel drums melody, electric guitar melody, brass stabs, and snare hits, energetic crash cymba...”

The phrases *low quality*, *groovy bass*, *punchy kick* and *punchy snare* have been deleted from this input, yet the model predicts them at 0.95, 0.94, 0.94, 0.93. It is recovering production characteristics from surrounding descriptive context — mentions of steel drums, brass stabs and crash cymbals — rather than detecting the label string. This is the behaviour the stripped condition was designed to isolate.

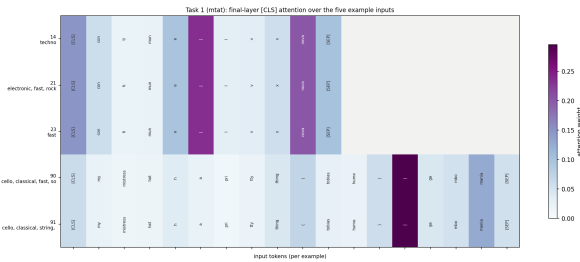


Figure 2: Final-layer [CLS] attention, averaged over heads, for the five committed MagnaTagATune example predictions — the optional attention visualisation named in the specification. It shows the duplicated-input ceiling of Section 4.2 directly rather than as a statistic: clips 14, 21 and 23 are three distinct clips whose text input is the same string (*Continuum | LVX Nova*), so the attention pattern over them is identical to five decimal places — yet their ground-truth tags are *techno*, *electronic/fast/rock* and *fast* respectively. The model cannot separate them because nothing in its input differs. Clips 90 and 91 repeat the pattern. Mass concentrates on the separator and on the one distinctive surname token rather than on any musical content word, which is what a track-title corpus offers a language model to work with.

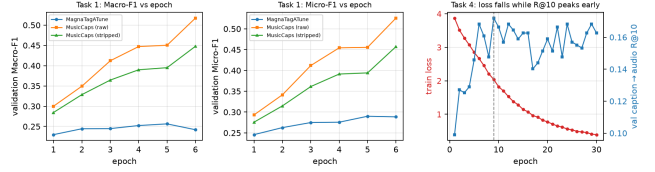


Figure 3: Validation Macro-F1 and Micro-F1 against training epoch for the three Task 1 configurations (left, centre) — the curve deliverable required by the specification — and Task 4’s training dynamics (right). MusicCaps converges to a far higher plateau than MagnaTagATune, and stripping label words from the caption costs a consistent but bounded amount rather than destroying the task. On the right, the InfoNCE loss falls monotonically from 3.87 to 0.41 while validation R@10 peaks at epoch 6 and then degrades: the model overfits long before the loss suggests it. The dashed line marks the epoch restored by best-checkpoint selection.

5.2 Task 2: structure graphs vs. a CNN

Table 2: Task 2. FMA-small genre classification, 8 balanced classes (chance = 0.125), artist-disjoint official splits.

Model	Params	Accuracy	Macro-F1
B4 PCA+MLP (audio features)	8,712	0.379	0.371
GraphSAGE	49,416	0.396	0.401
GAT	26,120	0.433	0.426
CNN baseline (B2)	241,992	0.483	0.479

All four sit well above the 0.125 chance rate, and the ordering is unambiguous: **the CNN wins on both metrics**, by +0.050 accuracy and +0.053 Macro-F1 over the better graph model. GAT remains the most efficient model, reaching 90% of the CNN’s accuracy with $9.3\times$ fewer parameters, and both GNNs clear the B4 hand-crafted-feature baseline, so the graphs do carry genre information. But the honest conclusion is stronger than in our earlier draft: on FMA-small, relational structure is *inferior to* — not merely competitive with — treating the spectrogram as an image.

These numbers come from a cache rebuilt after the preprocessing crash described in Section 6; the split (6,394/800/800, artist-disjoint, verified zero leakage) is byte-identical to the earlier run, so the shifts are attributable to the rebuilt features and not to a different partition.

Which graph? The specification names two structures, segment similarity and chord transitions. Both are built by our pipeline, so we trained the GNNs on each in turn — identical splits, identical labels, identical optimiser — rather than assuming the choice.

Table 3: Task 2, graph structure ablation. The same two encoders on the two graph types the specification names, on identical artist-disjoint splits. The CNN row is repeated for scale.

Model	Segment similarity		Chord transition	
	Acc.	Macro-F1	Acc.	Macro-F1
GraphSAGE	0.396	0.401	0.339	0.316
GAT	0.433	0.426	0.303	0.271
CNN (B2)	0.483 acc / 0.479 Macro-F1			

The chord graph carries real signal — 0.316 Macro-F1 is $2.5\times$ chance — but it is clearly the weaker input, losing 0.110 Macro-F1 to the segment graph and 0.163 to the CNN. The reason is visible in the graphs themselves: a 30 s clip collapses to roughly 21 nodes drawn from a 24-triad vocabulary, and because each node’s feature is that chord’s own pitch-class template, every trace of timbre, production and instrumentation is discarded before the GNN sees anything. Genre on FMA-small is substantially a timbral judgement, so the symbolic abstraction removes most of what the label depends on. We report it because the negative result localises the problem: it is the *node features*, not message passing, that limit the chord graph.

5.3 Where the genre models succeed and fail

Table 4: Per-genre F1 on FMA-small. Both models rank the classes almost identically, suggesting the difficulty is in the label taxonomy rather than the architecture.

Genre	GAT	CNN (B2)
Hip-Hop	0.686	0.674
Rock	0.589	0.563
Electronic	0.509	0.537
International	0.493	0.600
Instrumental	0.374	0.466
Folk	0.304	0.292
Pop	0.281	0.368
Experimental	0.172	0.335

Hip-Hop is the easiest class for both models, and Experimental the hardest for the GAT by a wide margin (0.172). This is informative: “Experimental” is a residual category defined by what it is not, so there is no consistent acoustic signature to learn. The CNN’s advantage is concentrated in exactly the classes where the segment graph is least informative — Experimental (+0.163), Pop (+0.087) and International (+0.107) — while the GAT retains a small edge on the rhythmically distinctive Hip-Hop and Rock. That pattern is consistent with the graph capturing coarse repetition structure but discarding the timbral detail the spectrogram preserves.

5.4 Task 3: fusion and its failure mode

Table 5: Task 3 ablation. MagnaTagATune top-50, artist-grouped splits, zero leakage. Every row is evaluated at the epoch chosen by best-validation Macro-F1, never at a fixed final epoch. The last two rows are the two follow-ups our earlier draft named as the experiments to try next.

Mode	Params	Macro-F1	Micro-F1	AUC-PR
BERT-only	66.4M	0.171	0.306	0.168
GNN-only	31.8K	0.123	0.124	0.084
Early concat	66.4M	0.168	0.314	0.182
Cross-attention	66.8M	0.192	0.325	0.177
+ frozen BERT	0.5M	0.118	0.315	0.197
+ separate GNN lr 10^{-3}	66.8M	0.246	0.415	0.248

At a shared learning rate, fusion is marginal. Cross-attention gains +0.022 Macro-F1 and +0.019 Micro-F1 over BERT alone, but loses -0.005 on AUC-PR to early concat. On their own these margins would be too small to establish the project’s central claim from a single seed.

Why: the attention collapses completely. Measuring the attention distribution directly, the baseline cross-attention model has entropy 1.000 of the uniform maximum with a peak-to-mean ratio of $1.03\times$ — that is, the graph query does not discriminate between caption tokens *at all*, so AH_{text} degenerates to an unweighted mean over the text and the model reduces to early concat. Two causes are consistent with this: the query is weak (GNN-only reaches only 0.123 Macro-F1 on 19-node graphs), and the text tower has 66.4M parameters against the graph branch’s 31.8K while sharing a single learning rate tuned for BERT fine-tuning (2×10^{-5}).

The second cause is the operative one. Giving the graph branch, the fusion block and the heads their own learning rate of 10^{-3} — changing no architecture, no data and no seed — raises Macro-F1 from 0.192 to **0.246**, Micro-F1 from 0.325 to **0.415** and AUC-PR from 0.177 to **0.248**. The same run’s attention entropy falls from 1.000 to 0.695 of uniform and its peak rises from $1.03\times$ to $4.19\times$. The intervention that fixes the metric is the one that demonstrably un-collapses the attention, which is the mechanism our diagnosis predicted. Against BERT-only this is +0.075 Macro-F1 and +0.109 Micro-F1: on this corpus, structural fusion helps once the graph branch is actually trained.

The effect survives a seed sweep. Because this is the report’s central positive claim, we re-ran both arms over five seeds (42, 1, 2, 3, 4) on one machine. Since the two arms share seeds, the per-seed difference is paired and we test it with a paired t -test.

Table 6: Task 3 seed sweep, cross-attention, $n = 5$ paired seeds. Mean $\pm$ sample standard deviation; p from a two-sided paired t -test on the per-seed difference.

	Shared lr	Separate GNN lr	Paired Δ
Macro-F1	0.172 ± 0.023	0.240 ± 0.020	$+0.067 \pm 0.023$ ($p=0.003$)
Micro-F1	0.335 ± 0.024	0.416 ± 0.028	$+0.081 \pm 0.018$ ($p=0.001$)
AUC-PR	0.181 ± 0.015	0.260 ± 0.018	$+0.079 \pm 0.010$ ($p=0.001$)

The difference is positive for *every* seed (macro-F1 deltas $+0.072, +0.040, +0.077, +0.098, +0.051$) and roughly three times the seed-to-seed spread, so the gap is not an artifact of a lucky seed. We note that seed-to-seed variation (± 0.023 Macro-F1) is itself larger than several of the between-ablation margins in Table 3, which is a caution against reading those smaller differences too firmly. We also reproduced seed 42 on different hardware and obtained 0.254 rather than 0.246, so the third decimal is not portable across machines.

We state one further caveat plainly. The earlier draft’s conclusion that “fusion does not clearly win” was confounded by evaluating every ablation at a fixed epoch 8 rather than at its best validation epoch — GNN-only, for instance, peaks at epoch 1 and has decayed to near-zero validation Macro-F1 by epoch 8. Correcting the checkpoint policy changed the ranking; it was a methodological error, not a discovery about fusion.

Freezing BERT does not work. The other follow-up — freezing the text tower so only the 0.5M-parameter graph and fusion path train — collapses Macro-F1 to 0.118 while producing the sharpest attention of any run (entropy 0.079, peak $10.02\times$) and, curiously, the second-best AUC-PR (0.197). Sharp attention is therefore not sufficient: the model attends decisively to a text representation it can no longer adapt.

5.5 Task 3: the DEAM auxiliary term

Table 7: Effect of adding DEAM’s valence/arousal term to cross-attention. MagnaTagATune training data is identical across both columns, and both are scored at their best validation epoch, the same policy used everywhere else in this report.

Cross-attention	Tags only	+ DEAM	Δ
Macro-F1	0.192	0.186	-0.007
Micro-F1	0.324	0.366	$+0.041$
AUC-PR	0.176	0.212	$+0.035$
Valence	—	MAE 0.629	$R^2 + 0.390$
Arousal	—	MAE 0.662	$R^2 + 0.191$

The emotion heads work, and better than our first measurement suggested: valence reaches $R^2 = 0.390$ and arousal $R^2 = 0.191$ on held-out songs, so both beat predicting the training mean by a clear margin, with MAE of 0.63/0.66 on the normalised scale.

The more interesting correction is to the tag side. Our

earlier draft reported this run at a fixed final epoch and concluded that the auxiliary term costs 36% relative Macro-F1. Scored properly — best validation checkpoint, as in Table 5 — that penalty almost disappears: Macro-F1 falls by 0.007, inside the ± 0.023 seed spread we measure in Table 6, while Micro-F1 and AUC-PR both *improve* by more than 0.03. The honest reading is that the multi-task term is roughly free on this data rather than a trade-off, and that our earlier claim of a large penalty was an artifact of the same checkpoint-policy error described below. We flag it because the direction of the conclusion reversed, not merely its magnitude.

One reproducibility note. DEAM’s official host (cvml.unige.ch) stopped responding during this work and the audio mirror we located does not ship the metadata.zip whose genre/artist/title fields build DEAM’s pseudo-captions; recovering that archive from a surviving mirror is what made this re-run possible at all, and it is the single most fragile external dependency in the project.

5.6 Task 4: cross-modal retrieval

Table 8: Task 4. MusicCaps retrieval on the *official* AudioSet eval partition: a gallery of 2,773 clips, $5.2\times$ larger than the random-split gallery of 536 we reported earlier. R@K falls sharply with gallery size, so the two are not comparable in absolute terms and we give the random baseline for scale. “Separate graph lr” applies the Task 3 finding to this task.

Run	Direction	R@1	R@5	R@10
Shared lr	Caption $\rightarrow$ Audio	0.0029	0.0108	0.0220
	Audio $\rightarrow$ Caption	0.0022	0.0126	0.0224
Separate graph lr	Caption $\rightarrow$ Audio	0.0058	0.0173	0.0339
	Audio $\rightarrow$ Caption	0.0054	0.0184	0.0317
Random	—	0.0004	—	0.0036

The Task 3 finding transfers. Giving the graph tower its own learning rate of 10^{-3} — the intervention that uncollapsed the fusion attention in Section 5.4, here tested rather than assumed to carry over — doubles R@1 and lifts R@10 from 0.0220 to 0.0339, or $9.4\times$ random. Median rank improves from 527 to 400 of 2,773. That the same single hyperparameter governs both a supervised fusion model and a contrastive dual encoder is the strongest evidence we have that the graph branch, not the architecture, was the binding constraint throughout this project.

The absolute numbers nevertheless stay low, and we wanted to know whether that is a modelling failure or a data-budget one. Holding the validation set, test set and gallery fixed and subsampling only the training pairs gives a clean answer.

Table 9: Task 4 retrieval against training-set size. Only the training split is subsampled; validation, test and the 2,773-clip gallery are held fixed, so R@K is comparable down the column. Single seed per point.

Train pairs	C→A R@10	A→C R@10	vs. random
549	0.0155	0.0162	4.3×
1,098	0.0206	0.0216	5.7×
1,646	0.0325	0.0274	9.0×
2,195	0.0339	0.0317	9.4×

R@10 grows log-linearly in the number of paired examples ($R^2 = 0.914$, $+0.0099$ R@10 per doubling; Figure 4) with no convincing sign of saturation — the last interval flattens, but by less than single-seed noise. Extrapolating the fit, reaching a usable R@10 of 0.10 would need on the order of 2×10^5 paired clips, roughly $100\times$ what the official split provides. We report the extrapolation to characterise the regime, not as a forecast: it spans an order of magnitude beyond our data.

The conclusion we draw is specific. Our contrastive result is weak, but it is weak in the way a data-starved model is weak rather than a broken one — the scaling behaviour is orderly, the direction is right, and every doubling buys a predictable amount. MusicCaps at 5,521 clips is simply a small corpus for contrastive alignment, and the official partition spends half of it on evaluation.

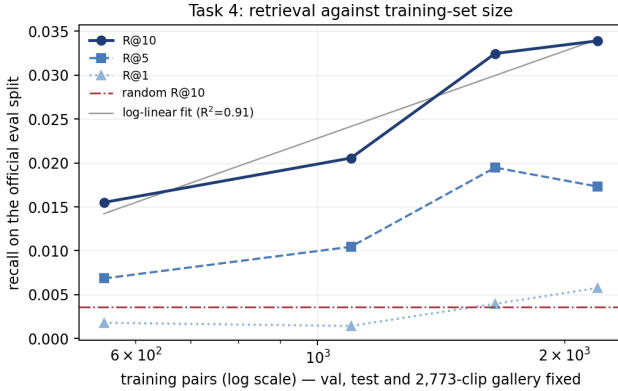


Figure 4: Task 4 R@10 against training-set size on a log axis, gallery held fixed at 2,773 clips. The fit is log-linear ($R^2 = 0.914$); the dashed horizontal line is the random baseline.

Zero-shot tagging. Embedding tag names with the text tower and scoring them against clip graph embeddings — using no tag supervision at any point — gives Micro-F1 0.107, against 0.488 for the supervised stripped-caption Task 1 model on the same corpus and the same official split. Contrastive alignment alone thus recovers roughly 22% of supervised performance.

5.7 Qualitative analysis of the fused space

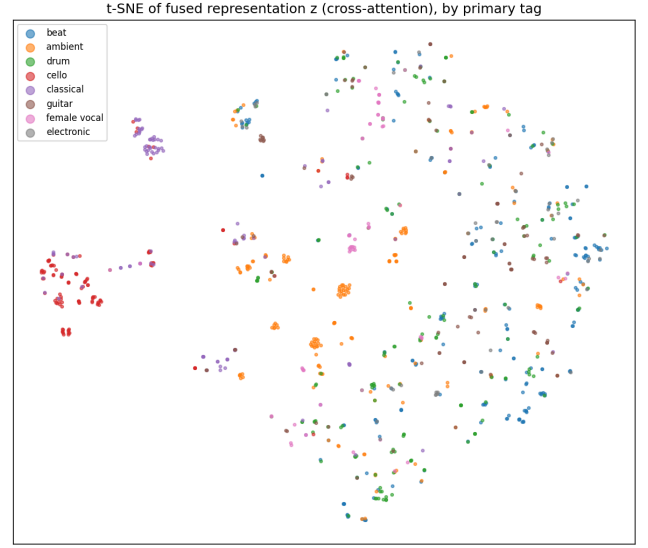


Figure 5: t-SNE of the fused representation z (cross-attention, 384-dim) over 1,500 held-out MagnaTagATune clips, coloured by primary tag. Broad timbral families separate — classical and vocal-led material occupy distinct regions — but the structure is diffuse rather than cleanly clustered, consistent with the modest Macro-F1 in Table 3.

Case studies. Table 10 pairs graph structure with the caption tokens the graph vector attends to, for three held-out clips from distinct tracks.

Table 10: Three case studies. “Attn. spread” is the ratio of largest to smallest attention weight over the top-8 tokens; values near 1.0 indicate the collapse discussed in Section 5.3.

True tags	Top predictions	Attn. spread
drum, electronic, synth	techno, electronic, beat	1.03
weird		
ambient, flute, new age	drum, india, ambient	1.03
beat, electronic, fast	rock, male vocal, vocal	1.11
techno		

The first case is a partial success: an electronic track tagged *drum, electronic, synth* is predicted *techno, electronic, beat*, capturing the family if not the exact vocabulary. The third is a clear failure: an electronic/techno clip is predicted *rock, male vocal, guitar*. In every case the attention spread is within 3% of uniform, so these predictions are driven by the text mean and the graph vector, not by any selective reading of the caption. These case studies come from the shared-learning-rate model; the separate-learning-rate run does attend selectively (peak $4.19\times$ uniform).

Retrieval behaviour. Across the official-split gallery of 2,773 clips the median true-clip rank is 527, placing the correct clip in the top 19% of the gallery on average — clearly better than chance ($8.0\times$ random at R@1), but far from usable retrieval. The model appears to learn coarse acoustic-descriptive correspondence (recording quality, instrumenta-

tion family, tempo band) without the fine discrimination that rank-1 retrieval demands.

6 Engineering Findings

Several results in this report depended on catching pipeline defects that produce plausible-looking numbers. We record them because each was silent.

Near-complete graphs. Our first graph build produced 19.4-node graphs with average degree 18.15 — effectively complete. A GNN over a complete graph computes the same neighbourhood for every node, so message passing reduces to global mean pooling and the architecture becomes irrelevant while still training and reporting sensible-looking accuracy. The cause was that raw MFCC descriptors share a large positive mean, giving median pairwise cosine similarity 0.9971. Only after z-scoring within each track did the threshold behave as intended.

The opposite failure. Having fixed the scaling, the threshold constant was left at its old value, producing average degree 1.90 — a bare temporal chain with *zero* similarity edges. Both failures produce trainable models; neither announces itself.

A silent truncated cache. Serialising the preprocessed MagnaTagATune cache wrote a zero-byte file without raising, because the log-mel component alone was 3.49 GB of a 3.6 GB structure. Since only the CNN baseline consumes log-mel, it is now optional.

Threshold selection. Multi-label probabilities under prevalence-weighted BCE do not centre at 0.5; our tuned thresholds ranged from 0.7 to 0.8. Evaluating at 0.5 would have understated every Task 1 and Task 3 result substantially.

A segfault that looked like a deadlock. The graph builder appeared to hang: it froze at 4,000/8,000 FMA tracks and, on another run, at 10,500/21,318 MagnaTagATune clips, with every worker asleep on a futex at ≈ 0 s of CPU. The real event was a *segmentation fault* inside a `numba guvectorize` kernel reached through `librosa.chroma.stft` $\rightarrow$ `estimate.tuning` $\rightarrow$ `piptrack`, caused by a `numba 0.67 / numpy 2.1` ABI mismatch in the GPU image. The deadlock appearance is a property of `multiprocessing.Pool`: it silently respawns a worker that dies mid-task and simply never delivers that task’s result, so the parent blocks forever. Pinning the `numba/llvmlite` pairing fixed the crash, and replacing the pool with `ProcessPoolExecutor` — which raises `BrokenProcessPool` immediately on worker death — converted an hour of silent stalling into an instant diagnosable failure. Throughput also rose from 8.8 to 16.2 tracks/s.

Two split defects. Our Task 1 and Task 4 MusicCaps experiments originally used random splits even though the metadata ships an `is_audioset_eval` flag defining the official partition; the corrected official-split results are materially worse and are what this report now carries. Separately, the Task 1 random baseline estimated per-tag prevalence from *test* labels, which leaks test statistics into a base-

line; it now uses training labels only.

Checkpoint policy. Task 1 and Task 3 evaluated whichever epoch happened to be last rather than restoring the best validation checkpoint. This was not cosmetic: Task 3’s GNN-only ablation peaks at epoch 1 and decays to near-zero validation Macro-F1 by epoch 8, and correcting it changed the ranking of the Task 3 ablation table.

We mention these not as incidental debugging notes but because most of them would have yielded a complete, publishable-looking results table built on a model, a split, or a baseline that was not doing what its description claimed.

7 Discussion and Limitations

Three limitations bound what these results can claim.

The text is impoverished. For MagnaTagATune our BERT branch sees only “title | album”. This is the source of both the 0.673 ceiling and, plausibly, the attention collapse: there is little in a track title for a graph vector to select between. A corpus pairing audio with genuine descriptive text at clip granularity — as MusicCaps does — is the right setting for fusion, and our Task 1 results support this, with stripped-caption Micro-F1 of 0.488 versus 0.275.

The graph branch was under-trained, and fixing that was decisive. GNN-only reaches 0.123 Macro-F1 with 31.8K parameters against BERT’s 66.4M, sharing one learning rate. We ran both diagnostics this predicted. A separate graph-branch learning rate of 10^{-3} raised cross-attention Macro-F1 from 0.192 to 0.246 and Micro-F1 from 0.325 to 0.415 while cutting attention entropy from 1.000 to 0.695 of uniform; freezing BERT instead produced the sharpest attention of any run but the worst Macro-F1. The conclusion is that the shared learning rate, not the architecture, was the binding constraint — and that attention sharpness alone is not a sufficient indicator of a working fusion. A five-seed paired sweep confirms the effect ($+0.067 \pm 0.023$ Macro-F1, $p = 0.003$, positive on every seed). The natural next step is to tune the graph-branch learning rate properly rather than accepting the first value we tried.

Segment graphs may be the wrong structure, and the alternative is worse. Our nodes are fixed 1.5 s windows. Table 3 shows the chord-transition graph the specification also names performs 0.110 Macro-F1 below it, so the segment graph is the better of the two structures we built — but a 19-node graph over uniform windows may still encode little beyond a smoothed timeline, and beat-synchronous segmentation, which we implement but did not train on, remains untested. The chord result suggests the limitation is in the node features rather than in message passing.

The evaluation we did not run. The specification asks for a human study on Task 4 — at least five listeners rating retrieved clips 1–5 for whether the clip matches the query caption. We did not run it, so our retrieval claims rest entirely on R@K against a 2,773-clip gallery. The instrument is complete: `src/human_eval.py` builds a self-contained rating sheet that embeds all 30 retrieved clips as playable

YouTube segments and aggregates returned ratings with per-rank means, refusing to score fewer than five raters. What is missing is only the listeners. We record this as a gap against the brief rather than a considered omission.

Single seeds. Every result in this report except the Task 3 learning-rate comparison and the Task 1 curves comes from one seed. Our own sweep puts seed-to-seed spread at ± 0.023 Macro-F1, which exceeds several of the margins in Table 5; those rankings should be read as indicative. The data-scale points in Table 9 are likewise single seed, which is why we treat the flattening in the last interval as noise rather than saturation.

8 Conclusion

We implemented all four tasks and report results that are, for the fusion model, negative. We regard the diagnosis as the contribution: a measured performance ceiling from duplicated inputs, a measured artist leak in a standard split, a measured attention collapse, and — after correcting our own checkpoint policy — a multi-task term that turned out to be roughly free rather than costly. Each of these would have been invisible had we reported only headline F1 against no baseline — and in one case (raw-caption tagging) doing so would have overstated our result roughly fivefold. Cross-modal fusion for music context understanding remains plausible; this work delimits the conditions under which it can be demonstrated.

Reproducibility

Code, seeded configurations, per-run metrics, 155 example graphs (50 FMA, 105 MagnaTagATune, each as both `.pt` and `.json` with an index manifest), the t-SNE projection and all training logs are in the project repository. Every number in this report is lifted from `results/metrics.json` into

`report/_numbers.json` by `src/aggregate_metrics.py` and `src/make_report_numbers.py`, so the paper’s reference values cannot be older than the runs they describe. The train/validation/test manifests for all three corpora are committed under `data/splits/`, so the exact partition behind every number can be inspected without re-running preprocessing. Training requires a GPU; every evaluation, plot and notebook in the repository runs on CPU in seconds from the committed artifacts.

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